A Donor-paired chamber estimand

Healthy reference donors

sample-level program activity

LA score RA score

Donor 1 LA — RA

Donor 2 LA — RA

Donor 3 LA — RA

⋮

n = 8 complete LA–RA donor pairs

↓

$$\Delta_{d,p} = S^{LA}_{d,p} - S^{RA}_{d,p}$$

S: sample-level program activity

one donor × one program × one paired contrast

Within-donor subtraction reduces donor-specific baseline variation.

Independent-sample pooling can mix donor and chamber effects.

Direction alignment and reference scaling → panel B

B Healthy-reference direction and magnitude across 30 programs

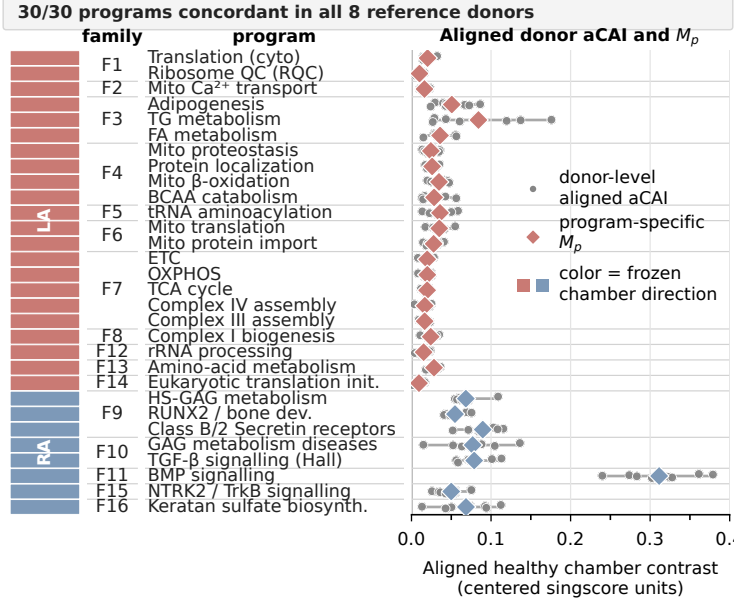
$aCAI_{d,p} = s_p \Delta_{d,p}$ $M_p = \text{mean}_{d \in H} |aCAI_{d,p}|$

$aCAIrel_{d,p} = aCAI_{d,p} / M_p$

All values are direction-aligned; color denotes the frozen LA/RA bias.

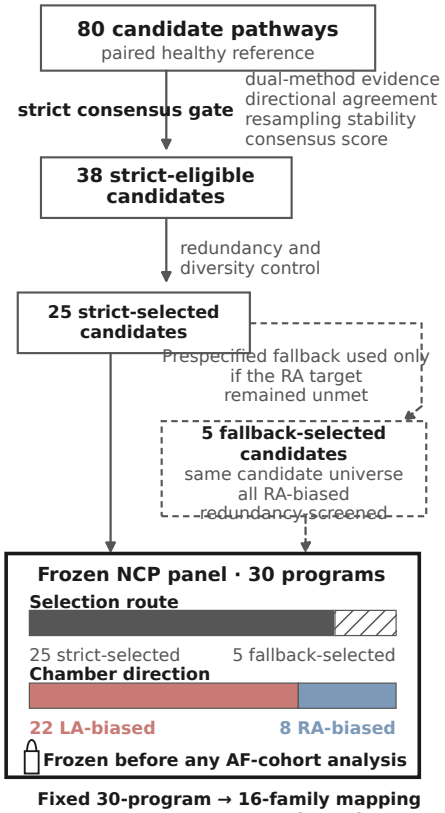
22 LA-biased | 8 RA-biased **median M_p = 0.0284**

30/30 programs concordant in all 8 reference donors



C Healthy-only selection and frozen deployment

GSE112339 paired healthy reference only · AF cohort data not used for selection



Fixed 30-program → 16-family mapping shown in D →

D Frozen NCP program-to-family topology

30 frozen programs → 16 families

11 LA-biased families | 5 RA-biased families

Fixed membership, direction and display order used in Figs 2–6

	ID	family	constituent programs	n
LA-biased	F1	**Translation / RQC**	Translation (cyto) · Ribosome QC (RQC)	**2**
	F2	**Mito Ca²⁺ transport**	Mito Ca²⁺ transport	**1**
	F3	**Adipo-lipid axis**	Adipogenesis · TG metabolism FA metabolism	**3**
	F4	**Mito proteostasis / FA-BCAA**	Mito proteostasis · Protein localization Mito β-oxidation · BCAA catabolism	**4**
	F5	**tRNA aminoacylation**	tRNA aminoacylation	**1**
	F6	**Mito translation / import**	Mito translation · Mito protein import	**2**
	F7	**OXPHOS / ETC**	ETC · OXPHOS TCA cycle · Complex IV assembly Complex III assembly	**5**
	F8	**Complex I biogenesis**	Complex I biogenesis	**1**
	F12	**rRNA processing**	rRNA processing	**1**
	F13	**Amino-acid metabolism**	Amino-acid metabolism	**1**
	F14	**Translation initiation**	Eukaryotic translation init.	**1**
RA-biased	F9	**HS-GAG / bone / GPCR**	HS-GAG metabolism · RUNX2 / bone dev. Class B/2 Secretin receptors	**3**
	F10	**GAG disease / TGF-β**	GAG metabolism diseases TGF-β signalling (Hall)	**2**
	F11	**BMP signalling**	BMP signalling	**1**
	F15	**NTRK2 / TrkB**	NTRK2 / TrkB signalling	**1**
	F16	**Keratan sulfate**	Keratan sulfate biosynth.	**1**

Primary transcriptomic family summaries use equal-weight program-first aggregation.